Socioeconomic Factors and ACT Performance in U.S. High Schools

Abstract

Educational performance varies substantially across U.S. high schools, and socioeconomic conditions may explain part of this variation. This study examined whether average school ACT scores, including SAT-equivalent scores, could be predicted using socioeconomic characteristics and school enrollment. School performance and socioeconomic data were obtained from EdGap, while school characteristics and enrollment were obtained from the National Center for Education Statistics Common Core of Data for the 2016–2017 school year. After data cleaning, restricting the sample to high schools, treating invalid values, and replacing a small number of missing predictor values, the final analysis included 7,227 schools from 20 states. Multiple linear regression was used to model ACT performance. The final model included unemployment rate, adult college attainment, free/reduced-price lunch participation, and log-transformed school enrollment. The model explained 63.1% of the variation in average ACT scores and had a mean absolute error of 1.14 ACT points. All four predictors were statistically significant. The results indicate that socioeconomic conditions provide substantial information about differences in school performance, although the limited geographic coverage and available predictors prevent the model from explaining all variation in ACT scores.

Introduction

Academic performance differs considerably across high schools, and these differences may be associated with the socioeconomic environments in which students live. Community unemployment, educational attainment, household income, and economic disadvantage can reflect differences in the resources and opportunities available to students. Examining these characteristics may therefore help explain variation in standardized test performance across schools.

This study uses school-level data from EdGap [1], which combines ACT or SAT-equivalent performance measures with socioeconomic characteristics associated with each school. The socioeconomic measures considered include unemployment rate, the percentage of adults with a college degree, the percentage of children living in married-couple families, median household income, and the percentage of students receiving free or reduced-price lunch.

School enrollment was added as an additional predictor using data from the National Center for Education Statistics (NCES) Common Core of Data [2]. Enrollment was included because school size may capture characteristics of the educational environment that are not represented directly by the socioeconomic measures.

The primary scientific aim of this study was to determine how well socioeconomic characteristics and school enrollment predict average ACT performance across U.S. high schools. A secondary goal was to determine which characteristics were most strongly associated with school performance and whether a more complex regression model produced a meaningful improvement over a simpler model.

Methods

Data

Three datasets from the 2016–2017 school year were used. The EdGap dataset provided school-level ACT or SAT-equivalent scores along with socioeconomic characteristics, including unemployment rate, adult college attainment, married-couple family prevalence, median household income, and free lunch participation [1].

School characteristics and enrollment were obtained from the NCES Common Core of Data Public School Universe Survey . The school information data identified school location, level, type, and charter status, while the enrollment data provided total student enrollment. Schools were matched across the datasets using the NCES school identifier. [2]

The analysis was restricted to schools classified as high schools because ACT and SAT performance is primarily relevant to secondary education. After matching the datasets and applying the inclusion criteria, 7,230 high schools remained before removal of observations with invalid or missing outcome values.

Data Preparation

The data were examined for impossible values, missing observations, and duplicated records. Negative ACT scores and negative free/reduced-price lunch proportions were considered invalid and treated as missing. Three schools with missing ACT scores were excluded because the response variable could not be recovered without directly estimating the outcome being modeled.

Missingness among the predictor variables was low, with less than 1% missing for each numerical predictor. Rather than discard schools that contained useful information in the remaining variables, missing numerical predictor values were replaced using iterative multivariable imputation.

School enrollment showed substantial right skew and several very large observations. Enrollment was therefore transformed using $log(1 + enrollment)$ which reduces the influence of extremely large schools while allowing schools with zero reported enrollment to remain mathematically defined. After data preparation, the final analysis contained 7,227 high schools.

The schools were not evenly distributed across the United States. The dataset included schools from 20 states, with the largest numbers from Texas, Ohio, Illinois, and Pennsylvania. Therefore, the sample should not be interpreted as nationally representative.

Statistical Analysis

Exploratory analysis was performed using scatterplots, boxplots, and Pearson correlation coefficients to evaluate relationships between ACT performance and the numerical predictors. Multiple ordinary least squares regression was then used to model average ACT score.

The initial regression model included unemployment rate, adult college attainment, median household income, free/reduced-price lunch percentage, and log-transformed enrollment. Because exploratory analysis suggested a curved relationship between household income and ACT performance, a second model included both linear and squared household-income terms.

A simpler final model was also evaluated using only unemployment rate, college attainment, free/reduced-price lunch percentage, and log-transformed enrollment. Regression coefficients were evaluated using two-sided significance tests with a significance level of (α=0.05). Model performance was assessed using (R^2) and mean absolute error (MAE). The simple and quadratic models were compared using an analysis of variance (ANOVA) for nested models.

Results

Exploratory Analysis

Socioeconomic characteristics showed substantial associations with average ACT performance. Free/reduced-price lunch participation had the strongest bivariate relationship with ACT score, with a correlation of approximately (r=-0.78). Thus, schools with higher proportions of students receiving free or reduced-price lunch tended to have substantially lower average ACT scores.

Unemployment rate was also negatively associated with ACT performance (r=-0.43). In contrast, adult college attainment and median household income were each positively associated with ACT performance (r=0.46). School enrollment had a weaker positive correlation with ACT score (r=0.18), and the correlation remained approximately 0.18 after the log transformation.

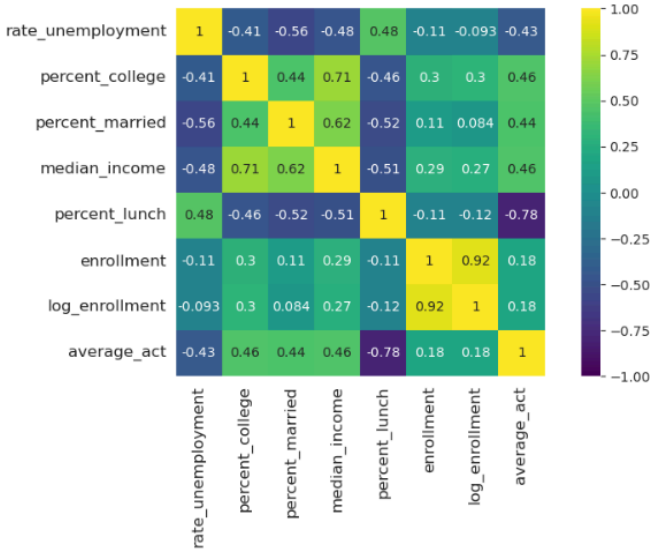


Figure 1. Correlation matrix of average ACT performance and numerical predictors.
Free/reduced-price lunch percentage had the strongest observed correlation with average ACT score (r=-0.78).

Several socioeconomic predictors were also correlated with one another. These relationships indicate that some socioeconomic variables contain overlapping information and reinforce the importance of evaluating predictors together in a multiple regression model rather than interpreting only their individual correlations.

Regression Analysis

In the initial multiple regression model, unemployment rate, college attainment, free/reduced-price lunch participation, and log-transformed enrollment were statistically significant predictors of average ACT score. Median household income was not statistically significant when modeled only as a linear term (p=0.211). The model explained approximately 63.1% of the variation in ACT scores.

Exploratory analysis indicated that the relationship between household income and ACT score was curved rather than strictly linear. Adding a squared household-income term increased (R^2) from approximately 0.631 to 0.641. Both the linear and quadratic income terms were statistically significant in the quadratic model. The model's MAE was 1.119 ACT points.

Although this represented a statistically significant improvement, the change in predictive accuracy was small. A simpler four-predictor model was therefore retained as the final model because it achieved nearly the same predictive performance while being easier to interpret.

Predictor	Coefficient	Standard Error	p-value
Intercept	21.532	0.172	<0.001
Unemployment rate	-2.258	0.372	<0.001
Adults with college degree	1.421	0.131	<0.001
Free/reduced-price lunch	-7.594	0.092	<0.001
Log-transformed enrollment	0.194	0.024	<0.001

Table 1. Final multiple linear regression model predicting average ACT score.

The final model explained 63.1% of the variation in average ACT scores (R^2=0.631). Holding the other predictors constant, a 10-percentage-point increase in unemployment rate was associated with an approximately 0.23-point decrease in predicted ACT score. A 10-percentage-point increase in adult college attainment was associated with an approximately 0.14-point increase in predicted ACT score. A 10-percentage-point increase in free/reduced-price lunch participation was associated with an approximately 0.76-point decrease in predicted ACT score.

Log-transformed enrollment also had a statistically significant positive coefficient, indicating that larger schools tended to have somewhat higher predicted ACT scores after controlling for the socioeconomic predictors. However, its bivariate relationship with ACT performance was considerably weaker than the relationship between ACT performance and free/reduced-price lunch participation.

The final model had a mean absolute error of 1.14 ACT points, meaning that the predicted school ACT score differed from the observed score by approximately 1.14 points on average.

The quadratic model produced an (R^2) of 0.641 and an MAE of 1.119, compared with an (R^2) of 0.631 and an MAE of 1.137 for the final simpler model. ANOVA indicated that adding the household-income terms significantly improved the model (F=100.02), (p<0.001). However, the approximately one-percentage-point increase in explained variation and 0.018-point reduction in MAE were small in practical terms. The four-predictor model was therefore retained as the final model.

Model Diagnostics

Residuals from the final model were generally distributed around zero across the range of predicted ACT scores. No strong systematic pattern of overprediction or underprediction was evident. However, several schools had relatively large positive or negative residuals, indicating that important determinants of school performance remain unaccounted for by the predictors included in the model.

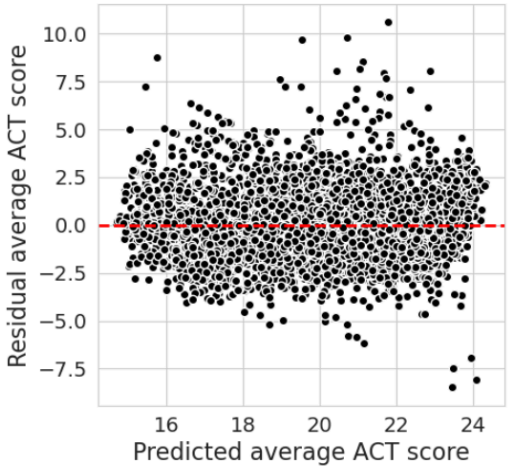


Figure 2. Residuals from the final multiple linear regression model versus predicted average ACT score.
Residuals were generally centered around zero, although several schools showed relatively large prediction errors.

Conclusion

This analysis examined whether school performance, measured by average ACT or SAT-equivalent scores, can be predicted using socioeconomic characteristics and school enrollment. The final regression model included unemployment rate, the percentage of adults with a college degree, the percentage of students receiving free or reduced-price lunch, and log-transformed school enrollment. Together, these variables explained approximately 63.1% of the variation in average ACT scores, with a mean absolute error of about 1.14 ACT points.

The results suggest that socioeconomic conditions are strongly associated with school performance. Higher unemployment rates and higher percentages of students receiving free or reduced-price lunch were associated with lower average ACT scores, while higher levels of college attainment were associated with higher scores. School enrollment, the additional predictor included in this project, also had a statistically significant positive relationship with ACT performance, although the relationship was weaker than those of the socioeconomic variables.

One limitation of this analysis is the coverage of the EdGap dataset. Although the final dataset contains more than 7,000 schools, the schools come from only 20 states and therefore do not fully represent all U.S. high schools. In addition, the available predictors capture only a limited set of socioeconomic and school characteristics. These limitations may explain why the final model could not achieve higher predictive accuracy, since other important factors affecting ACT performance are not included in the data.

Overall, the results show that socioeconomic factors provide substantial information for predicting school performance, but they cannot explain all differences in ACT scores. The relationships found in this analysis should also be interpreted as associations rather than causal effects.

References

[1] EdGap. *Educational Opportunity and Achievement Gap Data*. https://www.edgap.org/#5/37.892/-95.977

[2] National Center for Education Statistics. *Common Core of Data: Public School Universe Survey*. https://nces.ed.gov/ccd/pubschuniv.asp